

production in the United States in 1917 was the Curtiss flying boat. The US Army had about 50 obsolete combat aircraft. But if the United States lacked an aircraft industry, it did have an automobile industry that was using assembly-line techniques to transform output. Confident that if America could make cars it could make airplanes, Congress enthusiastically approved funds for the mass manufacture of aircraft. Optimistic plans saw the US equipping not only its own air service but those of Britain, France, and Italy. That is not how it worked out. Assembly-line methods proved hard to apply and differences between European and American standard measurements posed retooling problems. By the war's end, US factories had delivered only around 1,400 combat aircraft, mostly versions of the D.H.4 bomber.

Similar frustrations were experienced in engine manufacturing. Seven automobile manufacturers were contracted to make the Liberty aero-engine, but an original eight-cylinder design was declared obsolete before it went into production and had to be replaced by a heavier 12-cylinder model. Delays meant that only 1,300 Liberty engines had been delivered by June 1918. At 400hp they were the most powerful engines in the war – in fact far too powerful for existing airframes. American pilots who flew

MODERN WARFARE

During the Second Battle of the Marne in July 1918, a German biplane patrols the trenches as a British tank looms into view. The ubiquity of aircraft and tanks at this time was a foretaste of the type of mobile warfare that would predominate in World War II.

the Liberty-powered D.H.4s complained that if they ran the engine at full throttle it would shake the aircraft to pieces.

The final push

Germany launched its final offensive of the war, the *Kaiserschlacht*, on March 21, 1918. Using ground-attack aircraft in support of small groups of “shock troops,” the Germans punched holes in the Allied lines, ending the stalemate of trench warfare. In a repeat of 1914, Allied troops fell back toward Paris. The battle in the air was every bit as intense as on the ground. The Germans even

“What is the point of shooting down five out of 50 machines? The other 45 will... bomb as much as they want. The enemy's material superiority was... dooming us to failure.”

LIEUTENANT RUDOLF STARK

GERMAN PILOT DESCRIBING THE BATTLES OF 1918



threw their heavy Gotha bombers into the fray, attacking ammunition dumps behind the lines. The arrival of the superb Fokker D.VII at the end of April meant that German pilots had their best fighter of the entire war.

Yet with victory apparently in sight, the

German war effort began to crumble. By June, flying missions were being cut back due to lack of fuel. While the Allies were mostly able to replace their losses of aircraft and pilots, the Germans could not. American fliers were arriving in their thousands and their number would continue to increase. In July 1918, the Allies went on the offensive. In the air,

